

A Packing of 18 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 18 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.823104. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.814439, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 18 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.823104 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 18$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.814439, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 18/17 = 2.823529$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N18.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
141155231416105488209750498327154927977663052932267137459316208142793779
021668373572439781166653365541314358415830214848548687949317759585486846
9638884998238796474591516444008957212269342403418
```

denominator:

```
499999999999826349103145108231712730983828116524805921989611563269523943
749939744687584474767628752681978362925208583921914006645302347132625487
020507594134506865994968067605061356841921672003
```

B Coordinates

18 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N18.txt`.

```
-327973151761 565822447965 33522362740 -194615971530647658693 242516023705 613493420204 -399197957685-500456925533 113249231501 -16419532929 336447223025
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